O-ZONE TELCOM

WORKSHOP WITH CAPGEMINI CASE

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UNDERSTANDING THE CLIENT'S PROBLEM

Employee turnover follows a pattern — and patterns can be managed.

PROBLEM

- High employee turnover across operational roles.
- HR currently reacts only after resignation.
- No system exists to identify employees at risk.

BUSINESS IMPACT

- Recurrent recruitment and training costs.
- Loss of productivity during replacement periods.
- Loss of experience and team stability.
- Operational instability.

OBJECTIVE

- Give HR time to act before resignation
- Understand the drivers of attrition.
- Build a predictive early-warning model.
- Support data-driven HR decision-making

ANALYTICAL METHODOLOGY

How we turned HR data into actionable decisions

Exploratory Data Analysis

Identify when resignations happen.

Examination of employee profiles and work conditions to understand where attrition was concentrated

Feature simplification (PCA)

Reveal employee work situations.

We simplified the dataset by grouping related HR indicators into broader behavioral patterns, allowing clearer interpretation and more reliable modeling

Employee Segmentation (K-MEANS)

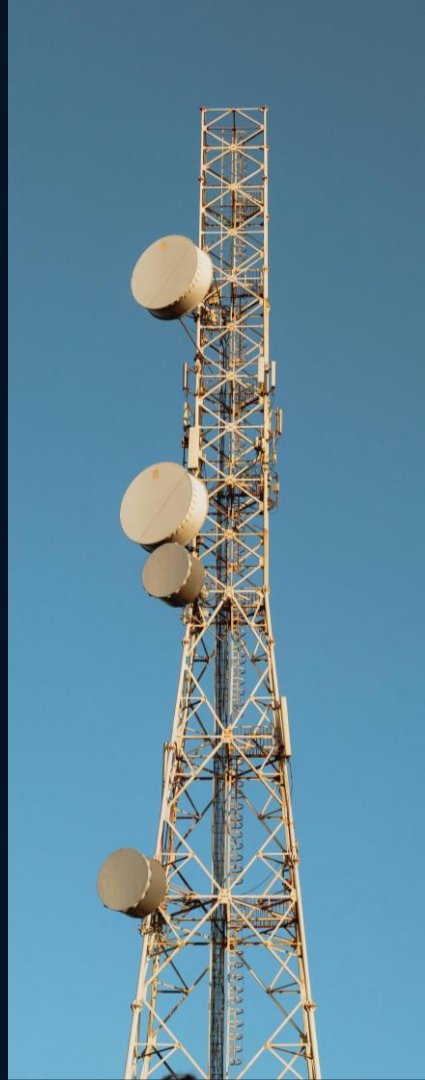
Different employee profiles emerge.

Employees were grouped into profiles with similar working conditions and satisfaction levels, revealing which types of employees are most exposed to attrition risk.

Predictive Model (Random Forest)

HR alerted before resignation.

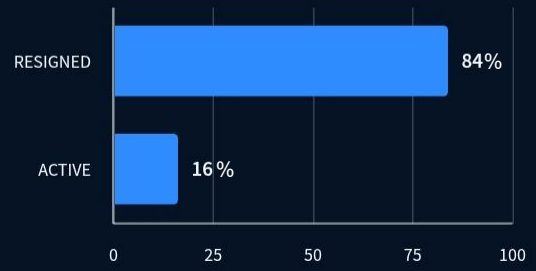
Finally, we built a predictive system that estimates the probability of resignation for each employee.



EXPLORATORY ANALYSIS

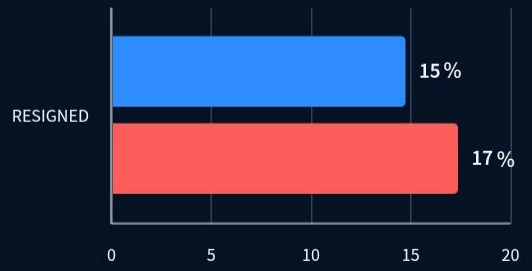
Turnover affects a specific type of employee

ATTRITION RATE 16.30% 1 IN 6 EMPLOYEES



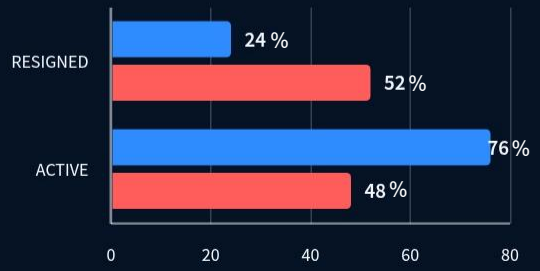
GENDER IS NOT THE PROBLEM.

Personal characteristics do not explain resignations.



OVERTIME IS THE STRONGEST WARNING SIGN

Working conditions drive resignations



EMPLOYEE SEGMENTATION ANALYSIS

The company does not have one workforce — it has different employee situations.

CHARACTERISTICS	EMPLOYEES TYPE A — NEW EMPLOYEES STILL ADAPTING TO THE COMPANY (~ 14%)	EMPLOYEES TYPE B — EMPLOYEES GROWING BUT UNDER PRESSURE (~ 61%)	EMPLOYEES TYPE C — STABLE EXPERIENCED EMPLOYEES (~ 25%)
ATTRIBUTION RATE	High (~20%)	High (~19%)	Low (~8%)
STAGE	Early organizational tenure	Career progression stage	Mature organizational tenure
TENURE	Recently hired	Several years in company	Long
MONTHLY INCOME	Low salary	Medium salary	High salary
JOB LEVEL	Entry / junior	Intermediate roles	Senior / managerial roles

MOST RESIGNATIONS HAPPEN BEFORE EMPLOYEES REACH STABILITY.

WORKFORCE STRUCTURE INSIGHT

- Employees follow a career progression path inside the company.
- Attrition occurs before reaching stable roles.
- The company loses employees during early career development.
- Turnover is structural, not random.

PREDICTIVE MODEL (HR EARLY-WARNING TOOL)

Alerts HR before employees resign.

ROC-AUC

0.81

RECALL

73%

Detects most future resignations (43 of 74 detected).

PRECISION

49%

Acceptable false alerts (better than missing a resignation).

TRAIN/TEST SPLIT

70/30

Tested on new employees

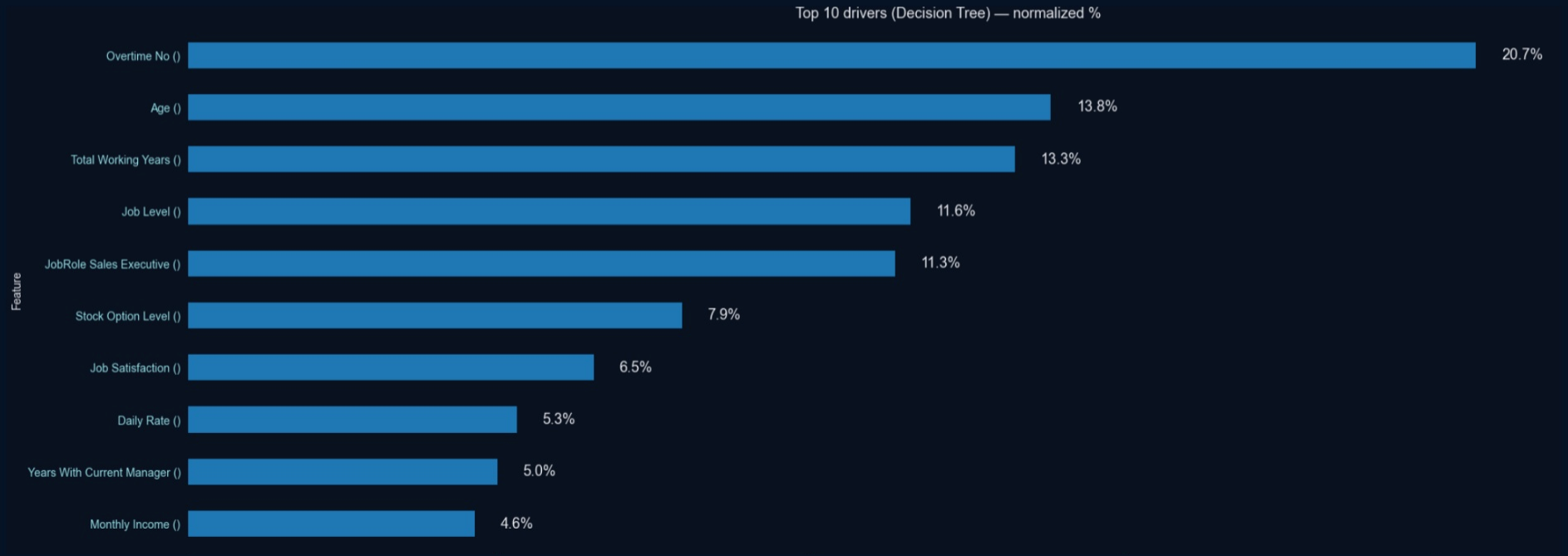
HOW HR USES IT:

- Monthly risk list.
- Each employee receives a risk score.
- HR can intervene before resignation occurs.

THE GOAL IS NOT TO PREDICT PEOPLE — BUT TO GIVE HR TIME TO RETAIN THEM.

KEY ATTRITION DRIVERS

Employees do not leave because of who they are — they leave because of their work situation.
Heavy workload + early career stage + limited growth = highest resignation risk



BUSINESS RECOMMENDATIONS

Because attrition is predictable, interventions can be targeted.

EARLY-CAREER RETENTION

- Mentoring during first 12 months.
- Career path clarity from day 1.
- Focus on first 12 months

COMPENSATION ADJUSTMENT

- Salary benchmarking for low-income roles.
- Retention bonus for high-risk employees.
- Transparent pay progression framework.

WORKLOAD CONTROL

- Prioritize overtime employees.
- Cap overtime for early-career staff.
- Automated overtime monitoring alerts
- Mandatory rest periods enforced by HR

MANAGERIAL CONTINUITY

- Reduce manager rotation in first 2 years.
- Leadership training for direct managers.
- Manager-employee matching for new hires.

CONCLUSION & TAKEAWAYS

Attrition is predictable and therefore preventable.

CONCLUSION

- Instead of general HR policies, the company the company should focus on:
  new employees
  overloaded roles
  early career support.
- The goal is not to predict people — the goal is to support them before they decide to leave.

KEY TAKEAWAYS

- 16.3% attrition → real cost problem
- Systemic cause, not gender
- 2 workforce profiles identified
- RF model · AUC 0.81.
- 4 targeted interventions.